

Multiple Choice (10 marks)

1. The best type of laser with which to do spectroscopy over a range of visible wavelengths is
 - (a) a dye laser
 - (b) a helium-neon laser
 - (c) an excimer laser
 - (d) a ruby laser
2. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
 - (a) Diode laser
 - (b) Dye laser
 - (c) Free-electron laser
 - (d) Gas laser
3. Electromagnetic radiation emitted from a nucleus is most likely to be in the form of
 - (a) gamma rays
 - (b) microwaves
 - (c) ultraviolet radiation
 - (d) visible light
4. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
5. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.

True / False (10 marks)

Answer True or False

6. True or False: The emission spectrum of the doubly ionized lithium atom Li^{++} has wavelengths decreased by a factor of 49 compared to hydrogen.
7. True or False: During a reversible thermodynamic process, the internal energy of the system always increases.
8. True or False: The frequency of the next higher harmonic for a closed organ pipe with a fundamental frequency of 131 Hz is 393 Hz.
9. True or False: In superconductors, the attraction between Cooper pairs is primarily due to interactions with the ionic lattice.
10. True or False: The eigenvalues of a Hermitian operator are always real numbers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between the kinetic energy of electrons and the diffraction pattern observed when they impinge on a crystal surface, specifically considering a lattice spacing of 0.4 nm. What energy level is necessary to observe this pattern, and why?
12. Discuss the implications of replacing the Sun with a black hole of the same mass on the orbits of the planets. How would gravitational forces and orbital mechanics remain unchanged?

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